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Introduction

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Super-resolution microscopy: a brief history and new avenues

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Super-resolution microscopy (SRM) is a fast-developing field that encompasses fluorescence imaging techniques with the capability to resolve objects below the classical diffraction limit of optical resolution. Acknowledged with the Nobel prize in 2014, numerous SRM methods have meanwhile evolved and are being widely applied in biomedical research, all with specific strengths and shortcomings. While some techniques are capable of nanometre-scale molecular resolution, others are geared towards volumetric three-dimensional multi-colour or fast live-cell imaging. In this editorial review, we pick on the latest trends in the field. We start with a brief historical overview of both conceptual and commercial developments. Next, we highlight important parameters for imaging successfully with a particular super-resolution modality. Finally, we discuss the importance of reproducibility and quality control and the significance of open-source tools in microscopy.

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1. A brief history of super-resolution microscopy

Optical fluorescence microscopy is a key method in modern biological and biomedical research. However, it has for the longest time suffered from the fundamental limitation of optical resolution imposed by the numerical aperture (NA) of the objective and the wavelength of light following fundamental laws of diffraction as described by Ernst Abbe [1]. Electron microscopy on the other hand, despite its ability to achieve orders of magnitude better resolution, has traditionally been hampered by the challenges in sample preparation and the difficulty in labeling and identifying specific molecules and structures, as well as a very low throughput. Hence physicists sought for ways to overcome the resolution barrier and bridge the gap between light and electron microscopy. Early works were confined to near-field scanning methods which had only very specialized and limited applicability in biology [2]. The first basic concepts to surpass the optical diffraction limit in far-field fluorescence microscopy were conceived in the early 1990s (see timeline in figure 1*a*). Along with optical sectioning through out-of-focus light rejection, slight improvements in the lateral resolution were already achieved by confocal microscopy [3]. Early super-resolution microscopy (SRM) developments initially addressed the inferior axial (z) resolution in far-field epifluorescence microscopy to achieve close to isotropic three-dimensional resolution by standing wave illumination [4] using opposing objectives in confocal (4Pi) [5,6] or widefield (I5 M) [7] microscopy. In the lateral direction, the theoretical foundations of stimulated emission depletion (STED) [8–10], structured illumination microscopy (SIM) [11–14] and single-molecule localization microscopy (SMLM) [15,16]. Yet it required the maturation of underlying laser, photo-detector and computer technology to trigger the 'Cambrian explosion' of practical implementations of SR techniques seen a few years later in the 2000s [17–23]. For SMLM, the *d*STORM variant, which works with conventional fluorophores, became most popular [24,25]. At this point, three major SRM techniques were established, i.e. STED, SIM and SMLM (now the widely accepted umbrella term for a raft of techniques that essentially follow the same principle and only distinguished themselves by the mechanism of stochastic switching of dye excitation states; see also table 1).

After 'overcoming' the fundamental limits of optical resolution, subsequent developments were mainly driven by the demands of biologists, that is to (i) enhance resolution in all spatial dimensions and/or enable volumetric imaging [26–35], (ii) improve temporal resolution and reduce photodamage at the expense of spatial resolution to enable live-cell SR imaging [36–51], and (iii) increase imaging depth and penetration for three-dimensional tissue imaging [52–57]. As a logical consequence, commercial implementation into turn-key instruments followed with a few years' delay (figure 1*b*), as did the widespread application as seen by the development of the number of publications (figure 1*c*).

The latest development push has come from combining SMLM with illumination principles of other SRM techniques, like 4Pi single-molecule switching (4Pi-SMS) [58], MINFLUX [59] and SIMFLUX type methods [60–63] to further increase either lateral or axial localization precision (figure 1*a*). Most recently, in random illumination microscopy (RIM), principles of SIM and SOFI are combined to enable 3D SRM in a more robust and user-friendly way that is less affected by aberrations [64]. In table 1, we provide an overview of major SRM techniques and their characteristics. For a more in-depth explanation of their functioning, we would like to refer to recent reviews [65–67]. Readers are also pointed to the following for further reference on this topic [68–72].

2. New avenues for super-resolution microscopy

No matter what leads to the development of a particular microscopy technique, whether it is the curiosity to build something new, to achieve the highest possible resolution, to gain new insights in natural, material or other sciences, to prove new physical phenomena or simply the joy of the technical interaction between man and machine—microscopy is almost always the result of many different disciplines working together. Nobel Prize-winning methods such as Cryo-EM, PALM or

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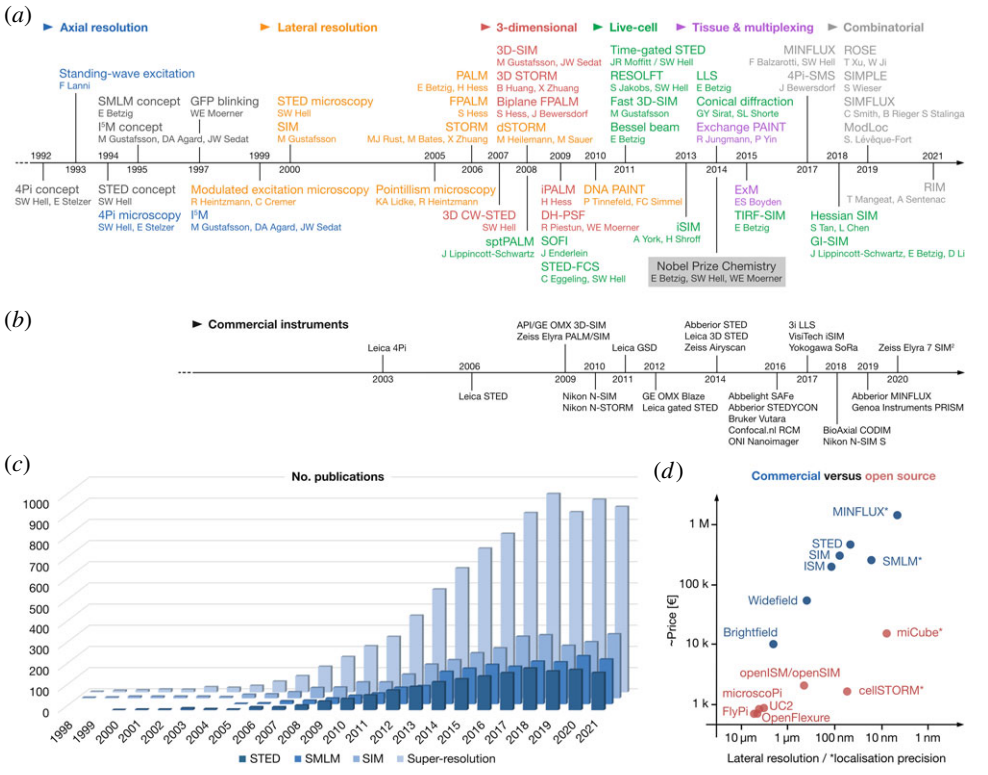


Figure 1. A brief history of far-field SRM. (*a*) Timelines highlighting crucial milestones in the development of SRM techniques (top) and the corresponding introduction of commercial turn-key super-resolution systems (bottom). For each SRM technique, the corresponding authors of the subsequent publication have been listed. (*c*) The number of publications extracted from the Web of Science database. (*d*) Correlation between the approximate purchase costs and the nominal lateral (*xy*) resolution or localization precision (SMLM-based systems) achievable under ideal conditions, for commercial and open-source microscopy systems, respectively. (Online version in colour.)

STED show that with the combination of new technology, the clever use of photophysical dye properties or computational tools, the optical resolution limit once described by Ernst Abbe can be circumvented.

Nonetheless, there is still plenty of room for ongoing and future improvements. Importantly, current progress is mainly geared towards promoting the reliability and applicability of these advanced techniques, rather than to further increasing resolution. On the instrument side, the wider-spread implementation of adaptive optics (AO) to correct (sample-induced) aberrations will benefit all the above-mentioned techniques to achieve their theoretical resolution in less ideal optical conditions, e.g. in thick cells and tissues [73–78].

Major advances are expected from the development of correlative microscopy approaches that seek to combine the strengths of different complementary imaging techniques [79–81]. For instance, recent advances in super-resolution cell imaging under cryo-conditions [82–84] and three-dimensional electron microscopy using advanced FIB-SEM [85] open up a potential pathway for developing new powerful three-dimensional CLEM workflows. Also, the consecutive application of SIM and single-molecule imaging techniques (SMLM, SPT) on the same widefield imaging platform holds the promise of getting the best of two worlds: single-molecule localization and contextual and structural three-dimensional information from SIM.

Major progress has been and continues to be provided by improvement in fluorescent dyes, probe design and labelling tools, such as in DNA-PAINT [86], Halo-tag JF dyes [87,88] and nanobody reagents [89]. These have been specifically designed with SRM in mind, to improve

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specificity and photon yield and live-cell imaging or, in combination with microfluidics or waveguide-based SRM imaging, enable multiplexing for future development [90–93].

Any progress on the instrument and labelling side will be accompanied by the massive utilization of bespoke Artificial Intelligence (AI) enhanced software solutions, implementing machine learning or neural networks to simplify and improve data post-processing/analysis [94,95] and bridging the gap to electron microscopy by correlative (cryo) super-resolution CLEM [96]. Accelerating data acquisition speed for fast, long-term imaging will be vital as the field moves towards live imaging [57,97–99]. A last important aspect is to continue to better understand the artefacts and limitations of individual SR approaches [100–102] and educate current and future users of SRM. A broader discussion of frequently asked questions in SRM can be found here [103–105].

3. Biological application of SRM - what have we learned?

Still, to this date, many SRM publications are generated in specialist physics/optics labs. These often display images of previously well-characterized macromolecules or biological structures, such as microtubules, nuclear pores, or actin filaments, as examples for biological applications. This led to the widespread impression that the applicability of SRM is somewhat limited, provoking the question 'What have we learned from SRM?' On closer examination, this is a misperception and SRM has become a genuine tool for discovery. However, what has also become clear over the past years is that, despite the promises of microscopy companies, most SRM is still not yet 'turn-key' in the same way conventional widefield and confocal microscopy is today. Particular expertise is required, not necessarily in how to 'press the right buttons' on any given commercial system, but for the typically more complex experimental design, the higher demands on the quality of sample preparations, the more delicate system calibration, and the complexity in data postprocessing and quantitative analyses. Therefore, the amount of time and commitment required to do SRM meticulously has been (and still is) a barrier for many biological and biomedical labs to move into this field. Recognizing this fact, the establishment of centralized core facilities has become a popular path to not only make advanced imaging systems available to a wider number of research labs but also to provide the expertise to run those systems effectively. Secondly, microscopy companies and developers are meanwhile turning their efforts into making SRM more accessible and reliable, which explains the success of 'soft super-resolution' methods, like rescan confocal, photon reassignment, Airy scan or iSIM [36,106,107], that come as easy-to-apply add-on features to standard confocal systems.

It is also important to realize that a biological discovery does not necessarily mean revealing a new structure. Rather than creating 'nice-looking' pictures (although that might be a pleasant side effect), the aim of an imaging experiment is rather to generate meaningful and reproducible quantitative data that helps to explain a biological phenomenon. Here, the ability to resolve events in time is as important as distinguishing objects or (macro)molecules in space as is their relationship to other molecules and structural features. High(er) throughput/content, along with elaborate data analyses, are becoming increasingly important for cutting-edge research involving SRM, and whatever the new findings, these need then be confirmed with orthogonal methods. Ideally, super-resolved images and data spark researchers to think differently about their particular biological problem and to question long-held assumptions.

With over a thousand SRM papers meanwhile published each year (figure 1*c*), it becomes increasingly difficult to pick out highlights of new discoveries, without doing injustice to many others. Good examples can be found in the field of chromatin and RNA biology, where single-cell 3D-SIM and SMLM studies lead to fundamentally new insights and models on how nucleosomes assemble higher-order structures and topological domains to define the functional modulus of genome organization [108–113], the crucial involvement of nanodomain formation in DNA repair [114], or how Xist RNA molecules spread during X-chromosome inactivation [115]. Larger macromolecular structures and enzyme complexes, such as synaptonemal complexes,

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Table 1. Overview of major SRM techniques and their characteristics.

Technique	Implementations & related methods	Principle / detection	Response	Diffraction -unlimited	Resolution	Merits	Limitations	Deconvolution
STED	2D/3D STED Time-gated STED DyMIN STED Tau-STED STED-FCS	Point scanning + PSF shaping Photo-detector	Non-linear	Yes	Structural resolution typically 40-60 nm (xy) and ~600 nm (z) for 2D STED, or ~100 nm (xy) and ~250 nm (z) in 3D STED; excitation/depletion power-dependent	- Excellent lateral resolution increase for sparse/isolated structures - Routine 2-colour imaging - No postprocessing required - High penetration depth - Variable resolution increase	- Low dynamic range - Sensitivity to out-of-focus background -> Less good for dense 3D features - High power density -> phototoxicity - Low imaging speed per field of view - Tricky to image more than 2 colours - Trade-off between z-resolution increase and xy-resolution in 3D STED	Routinely applied to improve signal to noise
SMLM	2D/3D/TIRF/HILO STORM dSTORM PALM DNA-PAINT	Widefield + temporal separation of stochastic emission (EM)CCD/sCMOS	Non-linear	Yes	Localisation precision typically 10-20 nm (xy); dependent on detected photons Structural resolution at least 2-fold lower; dependent on labelling-density	- Single-molecule sensitivity - Highest potential resolution	- Over-representation of sparse background, under-representation of very dense features - Typically, no more than 2-colour imaging (except Exchange PAINT) - Limited z-range in 3D SMLM - Very slow -> low throughput	Not typically applied
SOFI	SRRF ESI	Widefield + fluctuation autocorrelation (EM)CCD/sCMOS	Non-linear	Yes	Typically 100-150 nm (xy)	- No molecule identification needed - Enables live cell imaging	- Only moderate lateral resolution increase - SRRF particularly suited for fibres	Not typically applied
SIM[a]	TIRF-SIM GI-SIM 2/3-beam (2D/3D SIM) 5-beam Lattice SIM RIM	Widefield + moiré interference & frequency shifting (EM)CCD/sCMOS	Linear	No	Structural resolution max. 2-fold better than the diffraction limit, typically ~110 nm (xy) and ~320 nm (z) (1.4 NA, 488 nm excitation) NA- and modulation contrast-dependent	- High sensitivity - Rel. high speed per field of view - Efficient optical sectioning with much-increased contrast - Works with rel. dense features - Volumetric live-cell imaging - Routine 3/4-colour imaging	- Sensitivity to aberrations - Best results with measured OTFs and regular calibration - Resolution increase wavelength-dependent	Linear deconvolution (Wiener) typically applied Zeiss SIM² adds non-linear iterative deconvolution to further improve resolution
ISM	Re-Scan Airyscan iSIM	Single- or multi-point scanning Photo-detector (array) or sCMOS	Linear	No	Structural resolution 1.4-1.7-fold better than diffraction limit ("soft super-resolution")	- Easy applicability - Simple add-on to confocal system - Robust against artefacts	- Less efficient high-frequency transfer than interference-based SIM - Less contrast improvement than interference-based SIM	Non-linear iterative deconvolution typically applied to improve resolution
LLS	Bessel beam	Lattice light-sheet & mono-dimensional SIM sCMOS	Linear	No	Structural resolution 250 x 250 x 200 nm	- Highest temporal resolution per volume - Low phototoxicity - Fast & long-term volumetric live cell imaging - Near isotropic resolution	- Sub-diffraction resolution only in one dimension	Richardson-Lucy deconvolution typically applied

[a]Linear super-resolution SIM, as opposed to conventional resolution optical sectioning SIM (OS-SIM, e.g. Zeiss Apotome), or higher-resolving saturated SIM (SSIM), non-linear SIM (NL-SIM), and 3D pRESOLFT.

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centrosomes, kinetochores, DNA repair complexes, cytoskeleton, subcellular organelles, etc. lend themselves particularly well to super-resolution studies (for recent reviews see [65,116]).

After more than one decade of development, SRM has shown that different modalities have different biological application areas and specific sweet spots of individual methods are complementary to each other. However, there is still a prevalent lack of understanding of the general benefits and limitations of one method over the other, that goes beyond comparing nominal resolution numbers (see table 1). This applies e.g. to the crucial trade-offs in SRM often depicted as 'magical tetrahedron' of spatial resolution, temporal resolution, photodamage, and imaging depth. Beyond this, there are further less-known trade-offs. For instance, in SMLM, the ability to localize molecules with high precision does not necessarily enable the ability to visualize and resolve macromolecular structures with enough sampling density. Increasing precision can come at the cost of decreased probability to detect localizations [69]. Moreover, both SMLM and STED employ non-linear excitation/detection to become diffraction-unlimited which comes at the price of undercounting fluorescent signals/molecules in some areas of the sample while overestimating them in others. In contrast, in linear SIM relative intensities between features with fewer or higher labeling densities are retained, allowing valid intensity quantifications. Yet this advantage gets lost when non-linear iterative three-dimensional deconvolution is added on top, which renders the data useless for particular analyses as well as generating oversharpening artifacts.

Besides biological research, SRM also has the potential to be applied in other fields, such as clinical diagnostics, e.g. using SIM through the eye lens to image the human retina with increased detail [117], or in food research using AO-assisted SMLM to investigate the characteristics of oil droplets in emulsions [118].

4. Open technology developments for super-resolution microscopy

Historically, SRM developments have not been open-source, e.g. due to the demand for commercialization. However, more recently the philosophy of 'opening up' development projects to the entire research community and harnessing communal powers to accelerate progress, has gained traction. Projects like the Openflexure Microscope [119], the Fiji Image Analysis Software [120], the cellphone-based SMLM set-up 'cellSTORM' [121], the three-dimensionally printed modular toolbox UC2 [93,122]) demonstrate the importance of an open and active community for scientific discovery and collaboration. Users can use, modify and build on top of existing solutions [123–125] to acquire new data, analyse them and propose new theories or questions for future generations of scientists.

The sharing of data and resources has the immense advantage of enabling other researchers to reproduce the results or even recreate entire experiments. Hence, it picks up on a currently widely discussed debate: the reproduction crisis and the associated loss of society in scientific work. On top of open-source, 'frugal science' aims to make scientific instruments available at low to no costs. The core idea is to replace complex and usually expensive laboratory-grade devices with do-it-yourself or consumer-grade solutions. Wang *et al*. [122] succeeded in this in the manuscript 'UCsim2: 2D Structured Illumination Microscopy using UC2', in which the open-source three-dimensionally printed optics kit 'UC2' (You.See.Too.) is extended by super-resolution functionality using SIM and image scanning microscopy (ISM). The open-source documentation allows others to build a device themselves, e.g. for teaching purposes. A device that costs only 5000€ instead of one million also contributes to the fact that several experiments are carried out in many more places in the world (figure 1*d*). This allows laboratories in financially less well-equipped institutes to keep up with cutting-edge science and contribute to scientific progress. This approach also demonstrates the beauty of open-sourcing ideas.

In addition to the possibility of providing other scientists with the tools necessary for research to increase the reproducibility of scientific results, corresponding quality standards are of great importance for carrying out the experiments. Several initiatives such as the QUAREP-LiMi [126,127] recently proposed guidelines for good practice in (light) microscopic imaging and

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sample preparation. The standards developed together with the scientific community promise increased reproducibility across laboratories.

In this special issue on SRM, we pick up on these trends, show new advances in super-resolution imaging and also show how important it can be to focus not only on optical resolution but also on the reproducibility of scientific results, how quality standards and the creation of communities can help, and how scientific communities can be brought closer together so that tools can be developed that can be used to make breakthroughs.

Data accessibility. This article has no additional data.

Authors' contributions. K.P. and L.S.: conceptualization, writing—original draft, writing—review and editing; B.D.: writing—review and editing; R.H.: writing—review and editing. All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Competing interests. We declare we have no competing interests.

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References

1. Abbe E. 1873 Beitrage zur Theorie des Mikroskops und der mikroskopischen Wahrnehmung. *Arch. Mikr. Anat.* **9**, 413–468. (doi:10.1007/BF02956173)
2. Betzig E, Lewis A, Harootunian A, Isaacson M, Kratschmer E. 1986 Near field scanning optical microscopy (NSOM): development and biophysical applications. *Biophys. J.* **49**, 269–279. (doi:10.1016/S0006-3495(86)83640-2)
3. Pawley J (ed.) 2006 *Handbook of biological confocal microscopy*, vol. **236**. Berlin, Germany: Springer Science & Business Media.
4. Bailey B, Farkas DL, Taylor DL, Lanni F. 1993 Enhancement of axial resolution in fluorescence microscopy by standing-wave excitation. *Nature* **366**, 44–48. (doi:10.1038/366044a0)
5. Hell S, Stelzer EH. 1992 Properties of a 4Pi confocal fluorescence microscope. *JOSA A* **9**, 2159–2166. (doi:10.1364/JOSAA.9.002159)
6. Hell SW, Lindek S, Cremer C, Stelzer EH. 1994 Measurement of the 4Pi-confocal point spread function proves 75 nm axial resolution. *Appl. Phys. Lett.* **64**, 1335–1337. (doi:10.1063/1.111926)
7. Gustafsson MG, Agard DA, Sedat JW. 1995 Sevenfold improvement of axial resolution in 3D wide-field microscopy using two objective-lenses. In *Three-dimensional microscopy: image acquisition and processing II*, vol. 2412 (eds T Wilson, CJ Cogswell), pp. 147–156. Bellingham, WA: SPIE. (https://doi.org/10.1117/12.336833)
8. Okhonin V. 1991 A method of examination of sample microstructure. SU 1374922 A1. Priority 10.04.1986. (doi:10.13140/2.1.2588.1922)
9. Hell SW, Wichmann J. 1994 Breaking the diffraction resolution limit by stimulated emission: stimulated-emission-depletion fluorescence microscopy. *Opt. Lett.* **19**, 780–782. (doi:10.1364/OL.19.000780)
10. Baer SC. 1999 Method and apparatus for improving resolution in a scanned optical system. US Patent 5 866 911.
11. Heintzmann R, Cremer CG. 1999 Laterally modulated excitation microscopy: improvement of resolution by using a diffraction grating. In *Optical biopsies and microscopic techniques III*, vol. 3568 (eds IJ Bigio, H Schneckenburger, J Slavik, K Svanberg, PM Viallet), pp. 185–196. Bellingham, WA: SPIE.
12. Gustafsson MG. 2000 Surpassing the lateral resolution limit by a factor of two using structured illumination microscopy. *J. Microsc.* **198**, 82–87. (doi:10.1046/j.1365-2818.2000.00710.x)
13. Heintzmann R, Jovin TM, Cremer C. 2002 Saturated patterned excitation microscopy—a concept for optical resolution improvement. *JOSA A* **19**, 1599–1609. (doi:10.1364/JOSAA.19.001599)

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14. Gustafsson MG. 2005 Nonlinear structured-illumination microscopy: wide-field fluorescence imaging with theoretically unlimited resolution. *Proc. Natl Acad. Sci. USA* **102**, 13 081–13 086. (doi:10.1073/pnas.0406877102)
15. Betzig E. 1995 Proposed method for molecular optical imaging. *Opt. Lett.* **20**, 237–239. (doi:10.1364/OL.20.000237)
16. Dickson RM, Cubitt AB, Tsien RY, Moerner WE. 1997 On/off blinking and switching behaviour of single molecules of green fluorescent protein. *Nature* **388**, 355–358.
17. Klar TA, Jakobs S, Dyba M, Egner A, Hell SW. 2000 Fluorescence microscopy with diffraction resolution barrier broken by stimulated emission. *Proc. Natl Acad. Sci. USA* **97**, 8206–8210. (doi:10.1073/pnas.97.15.8206)
18. Lidke KA, Rieger B, Jovin TM, Heintzmann R. 2005 Superresolution by localization of quantum dots using blinking statistics. *Opt. Express* **13**, 7052–7062. (doi:10.1364/OPEX.13.007052)
19. Betzig E, Patterson GH, Sougrat R, Lindwasser OW, Olenych S, Bonifacino JS, Davidson MW, Lippincott-Schwartz J, Hess HF. 2006 Imaging intracellular fluorescent proteins at nanometer resolution. *Science* **313**, 1642–1645. (doi:10.1126/science.1127344)
20. Rust MJ, Bates M, Zhuang X. 2006 Stochastic optical reconstruction microscopy (storm) provides sub-diffraction-limit image resolution. *Nat. Methods* **3**, 793–796. (doi:10.1038/nmeth929)
21. Hess ST, Girirajan TP, Mason MD. 2006 Ultra-high resolution imaging by fluorescence photoactivation localization microscopy. *Biophys. J.* **91**, 4258–4272. (doi:10.1529/biophysj.106.091116)
22. Rego EH, Shao L, Macklin JJ, Winoto L, Johansson GA, Kamps-Hughes N, Davidson MW, Gustafsson MG. 2012 Nonlinear structured-illumination microscopy with a photoswitchable protein reveals cellular structures at 50-nm resolution. *Proc. Natl Acad. Sci. USA* **109**, E135–E143. (doi:10.1073/pnas.1107547108)
23. Jungmann R, Steinhauer C, Scheible M, Kuzyk A, Tinnefeld P, Simmel FC. 2010 Single-molecule kinetics and super-resolution microscopy by fluorescence imaging of transient binding on DNA origami. *Nano letters* **10**, 4756–4761.
24. Lemmer P *et al.* 2008 SPDM: light microscopy with single-molecule resolution at the nanoscale. *Appl. Phys. B* **93**, 1–12. (doi:10.1007/s00340-008-3152-x)
25. Heilemann M, Van De Linde S, Schüttpelz M, Kasper R, Seefeldt B, Mukherjee A, Tinnefeld P, Sauer M. 2008 Subdiffraction-resolution fluorescence imaging with conventional fluorescent probes. *Angew. Chem. Int. Ed.* **47**, 6172–6176. (doi:10.1002/anie.200802376)
26. Gustafsson MG, Shao L. Carlton PM. Wang CJR, Golubovskaya IN, Cande WZ, Agard DA, Sedat JW. 2008 Three-dimensional resolution doubling in wide-field fluorescence microscopy by structured illumination. *Biophys. J.* **94**, 4957–4970. (doi:10.1529/biophysj.107.120345)
27. Schermelleh L *et al.* 2008 Subdiffraction multicolor imaging of the nuclear periphery with 3d structured illumination microscopy. *Science* **320**, 1332–1336. (doi:10.1126/science.1156947)
28. Huang B, Jones SA, Brandenburg B, Zhuang X. 2008 Whole-cell 3D STORM reveals interactions between cellular structures with nanometer-scale resolution. *Nat. Methods* **5**, 1047–1052. (doi:10.1038/nmeth.1274)
29. Juette MF, Gould TJ, Lessard MD, Mlodzianoski MJ, Nagpure BS, Bennett BT, Hess ST. Bewersdorf J. 2008 Three-dimensional sub–100 nm resolution fluorescence microscopy of thick samples. *Nat. Methods* **5**, 527–529. (doi:10.1038/nmeth.1211)
30. Dertinger T, Colyer R, Iyer G, Weiss S, Enderlein J. 2009 Fast, background-free, 3D super-resolution optical fluctuation imaging (SOFI). *Proc. Natl Acad. Sci. USA* **106**, 22 287–22 292. (doi:10.1073/pnas.0907866106)
31. Hajj B, Wisniewski J, El Beheiry M, Chen J, Revyakin A, Wu C, Dahan M. 2014 Whole-cell, multicolor superresolution imaging using volumetric multifocus microscopy. *Proc. Natl Acad. Sci. USA* **111**, 17 480–17 485. (doi:10.1073/pnas.1412396111)
32. Abrahamsson S *et al.* 2017 Multifocus structured illumination microscopy for fast volumetric super-resolution imaging. *Biomed. Opt. Express* **8**, 4135–4140. (doi:10.1364/BOE.8.004135)

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33. Shtengel G *et al.* 2009 Interferometric fluorescent super-resolution microscopy resolves 3D cellular ultrastructure. *Proc. Natl Acad. Sci. USA* **106**, 3125–3130. (doi:10.1073/pnas.0813131106)
34. Pavani SR, Thompson MA, Biteen JS, Lord SJ, Liu N, Twieg RJ, Piestun R, Moerner WE. 2009 Three-dimensional, single-molecule fluorescence imaging beyond the diffraction limit by using a double-helix point spread function. *Proc. Natl Acad. Sci. USA* **106**, 2995–2999. (doi:10.1073/pnas.0900245106)
35. Willig KI, Harke B, Medda R, Hell SW. 2007 STED microscopy with continuous wave beams. *Nat. Methods* **4**, 915–918.
36. York AG, Parekh SH, Dalle Nogare D, Fischer RS, Temprine K, Mione M, Chitnis AB, Combs CA, Shroff H. 2012 Resolution doubling in live, multicellular organisms via multifocal structured illumination microscopy. *Nat. Methods* **9**, 749–754. (doi:10.1038/nmeth.2025)
37. Kner P, Chhun BB, Griffis ER, Winoto L, Gustafsson MG. 2009 Super-resolution video microscopy of live cells by structured illumination. *Nat. Methods* **6**, 339–342. (doi:10.1038/nmeth.1324)
38. Hirvonen LM, Wicker K, Mandula O, Heintzmann R. 2009 Structured illumination microscopy of a living cell. *Eur. Biophys. J.* **38**, 807–812. (doi:10.1007/s00249-009-0501-6)
39. Shao L, Kner P, Rego EH, Gustafsson MG. 2011 Super-resolution 3d microscopy of live whole cells using structured illumination. *Nat. Methods* **8**, 1044–1046. (doi:10.1038/nmeth.1734)
40. Li D *et al.* 2015 Extended-resolution structured illumination imaging of endocytic and cytoskeletal dynamics. *Science* **349**, aab3500. (doi:10.1126/science.aab3500)
41. Caron J, Fallet C, Tinevez JY, Moisan L, Braitbart LP, Sirat GY, Shorte SL. 2014 Conical diffraction illumination opens the way for low phototoxicity super-resolution imaging. *Cell adhesion & migration* **8**, 430–439.
42. Zhao W *et al.* 2021 Sparse deconvolution improves the resolution of live-cell super-resolution fluorescence microscopy. *Nature biotechnology* 1–12.
43. Manley S, Gillette JM, Patterson GH, Shroff H, Hess HF, Betzig E, Lippincott-Schwartz J. 2008 High-density mapping of single-molecule trajectories with photoactivated localization microscopy. *Nat. Methods* **5**, 155–157.
44. Gustafsson N, Culley S, Ashdown G, Owen DM, Pereira PM, Henriques R. 2016 Fast live-cell conventional fluorophore nanoscopy with ImageJ through super-resolution radial fluctuations. *Nature communications* **7**, 1–9.
45. Eggeling C *et al.* 2009 Direct observation of the nanoscale dynamics of membrane lipids in a living cell. *Nature* **457**, 1159–1162. (doi:10.1038/nature07596)
46. Caron J, Fallet C, Tinevez JY, Moisan L, Braitbart LP, Sirat GY, Shorte SL. 2014 Conical diffraction illumination opens the way for low phototoxicity super-resolution imaging. *Cell Adh. Migr.* **8**, 430–439. (doi:10.4161/cam.29358)
47. Moffitt JR, Osseforth C, Michaelis J. 2011 Time-gating improves the spatial resolution of STED microscopy. *Opt Express.* **19**, 4242–4254. (doi:10.1364/OE.19.004242)
48. Vicidomini G, Moneron G, Han KY, Westphal V, Ta H, Reuss M, Engelhardt J, Eggeling C, Hell SW. 2011 Sharper low-power STED nanoscopy by time gating. *Nat. Methods* **8**, 571–573. (doi:10.1038/nmeth.1624)
49. Grotjohann T, Testa I, Leutenegger M, Bock H, Urban NT, Lavoie-Cardinal F, Willig KI, Eggeling C, Jakobs S, Hell SW. 2011 Diffraction-unlimited all-optical imaging and writing with a photochromic GFP. *Nature* **478**, 204–208. (doi:10.1038/nature10497)
50. Guo Y *et al.* 2018 Visualizing intracellular organelle and cytoskeletal interactions at nanoscale resolution on millisecond timescales. *Cell* **175**, 1430–1442.
51. Huang X *et al.* 2018 Fast, long-term, super-resolution imaging with Hessian structured illumination microscopy. *Nature biotechnology* **36**, 451–459.
52. York AG, Chandris P, Dalle Nogare D, Head J, Wawrzusin P, Fischer RS, Chitnis A. Shroff H. 2013 Instant super-resolution imaging in live cells and embryos via analog image processing. *Nat. Methods* **10**, 1122–1126. (doi:10.1038/nmeth.2687)
53. Sheppard CJ. 2021 Structured illumination microscopy and image scanning microscopy: a review and comparison of imaging properties. *Phil. Trans. R. Soc. A* **379**, 20200154. (doi:10.1098/rsta.2020.0154)
54. Chen F, Tillberg PW, Boyden ES. 2015 Expansion microscopy. *Science* **347**, 543–548. (doi:10.1126/science.1260088)

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55. Ingaramo M, York AG, Wawrzusin P, Milberg O, Hong A, Weigert R, Shroff H, Patterson GH. 2014 Two-photon excitation improves multifocal structured illumination microscopy in thick scattering tissue. *Proc. Natl Acad. Sci. USA* **111**, 5254–5259. (doi:10.1073/pnas.1314447111)
56. Chen B-C *et al.* 2014 Lattice light-sheet microscopy: imaging molecules to embryos at high spatiotemporal resolution. *Science* **346**, 1257998. (doi:10.1126/science.1257998)
57. Liu T-L *et al.* 2018 Observing the cell in its native state: imaging subcellular dynamics in multicellular organisms. *Science* **360**, eaaq1392. (doi:10.1126/science.aaq1392)
58. Huang F *et al.* 2016 Ultra-high resolution 3D imaging of whole cells. *Cell* **166**, 1028–1040. (doi:10.1016/j.cell.2016.06.016)
59. Balzarotti F, Eilers Y, Gwosch KC, Gynnå AH, Westphal V, Stefani FD, Elf J, Hell SW. 2017 Nanometer resolution imaging and tracking of fluorescent molecules with minimal photon fluxes. *Science* **355**, 606–612. (doi:10.1126/science.aak9913)
60. Cnossen J, Hinsdale T, Thorsen RØ, Siemons M, Schueder F, Jungmann R, Smith CS, Rieger B, Stallinga S. 2020 Localization microscopy at doubled precision with patterned illumination. *Nat. Methods* **17**, 59–63. (doi:10.1038/s41592-019-0657-7)
61. Reymond L, Ziegler J, Knapp C, Wang F-C, Huser T, Ruprecht V, Wieser S. 2019 Simple: structured illumination based point localization estimator with enhanced precision. *Opt. Express* **27**, 24 578–24 590. (doi:10.1364/OE.27.024578)
62. Gu L, Li Y, Zhang S, Xue Y, Li W, Li D, Xu T, Ji W. 2019 Molecular resolution imaging by repetitive optical selective exposure. *Nat. Methods* **16**, 1114–1118. (doi:10.1038/s41592-019-0544-2)
63. Jouchet P, Cabriel C, Bourg N, Bardou M, Poüs C, Fort E, Lévêque-Fort S. 2021 Nanometric axial localization of single fluorescent molecules with modulated excitation. *Nat. Photonics* **15**, 297–304. (doi:10.1038/s41566-020-00749-9)
64. Mangeat T *et al.* 2021 Super-resolved live-cell imaging using Random Illumination Microscopy. *Cell Reports Methods* **1**, 100009.
65. Schermelleh L, Ferrand A, Huser T, Eggeling C, Sauer M, Biehlmaier O, Drummen GP. 2019 Super-resolution microscopy demystified. *Nat. Cell Biol.* **21**, 72–84. (doi:10.1038/s41556-018-0251-8)
66. Wu Y, Shroff H. 2018 Faster, sharper, and deeper: structured illumination microscopy for biological imaging. *Nat. Methods* **15**, 1011–1019. (doi:10.1038/s41592-018-0211-z)
67. Heintzmann R, Huser T. 2017 Super-resolution structured illumination microscopy. *Chem. Rev.* **117**, 13 890–13 908. (doi:10.1021/acs.chemrev.7b00218)
68. Reymond L, Huser T, Ruprecht V, Wieser S. 2020 Modulation-enhanced localization microscopy. *J. Phys.: Photonics* **2**, 041001. (doi:10.1088/2515-7647/ab9eac)
69. Prakash K. 2022 At the molecular resolution with MINFLUX? *Phil. Trans. R. Soc. A* **380**, 20200145. (doi:10.1098/rsta.2020.0145)
70. Cremer C, Birk U. 2022 Spatially modulated illumination microscopy: application perspectives in nuclear nanostructure analysis. *Phil. Trans. R. Soc. A* **380**, 20210152. (doi:10.1098/rsta.2021.0152)
71. Prakash K, Curd A. 2021 Assessment of 3D MINFLUX data for quantitative structural biology in cells. bioRxiv. See https://doi.org/10.1101/2021.08.10.455294.
72. Jouchet P, Poüs C, Fort E, Lévêque-Fort S. 2022 Time-modulated excitation for enhanced single-molecule localization microscopy. *Phil. Trans. R. Soc. A* **380**, 20200299. (doi:10.1098/rsta.2020.0299)
73. Débarre D, Botcherby EJ, Booth MJ, Wilson T. 2008 Adaptive optics for structured illumination microscopy. *Opt. Express* **16**, 9290–9305. (doi:10.1364/OE.16.009290)
74. Booth M, Andrade D, Burke D, Patton B, Zurauskas M. 2015 Aberrations and adaptive optics in super-resolution microscopy. *Microscopy* **64**, 251–261. (doi:10.1093/jmicro/dfv033)
75. Gould TJ, Burke D, Bewersdorf J, Booth MJ. 2012 Adaptive optics enables 3D STED microscopy in aberrating specimens. *Opt. Express* **20**, 20 998–21 009. (doi:10.1364/OE.20.020998)
76. Thomas B, Wolstenholme A, Chaudhari SN, Kipreos ET, Kner P. 2015 Enhanced resolution through thick tissue with structured illumination and adaptive optics. *J. Biomed. Opt.* **20**, 026006. (doi:10.1117/1.JBO.20.2.026006)

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77. Zheng W *et al.* 2017 Adaptive optics improves multiphoton super-resolution imaging. *Nat. Methods* **14**, 869–872. (doi:10.1038/nmeth.4337)
78. Zdankowski P, Trusiak M, McGloin D, Swedlow JR. 2019 Numerically enhanced stimulated emission depletion microscopy with adaptive optics for deep-tissue super-resolved imaging. *ACS Nano* **14**, 394–405. (doi:10.1021/acsnano.9b05891)
79. Liu F, Li Q, Jiang S, Zhou L, Zhang J, Zhang H. 2021 Two-photon structured illumination microscopy imaging using Fourier ptychography scheme. *Opt. Commun.* **489**, 126872. (doi:10.1016/j.optcom.2021.126872)
80. Keller PJ, Schmidt AD, Santella A, Khairy K, Bao Z, Wittbrodt J, Stelzer EH. 2010 Fast, high-contrast imaging of animal development with scanned light sheet-based structured illumination microscopy. *Nat. Methods* **7**, 637–642. (doi:10.1038/nmeth.1476)
81. Chang B-J, Meza VDP, Stelzer EH. 2017 csiLSFM combines light-sheet fluorescence microscopy and coherent structured illumination for a lateral resolution below 100 nm. *Proc. Natl Acad. Sci. USA* **114**, 4869–4874. (doi:10.1073/pnas.1609278114)
82. Kaufmann R, Schellenberger P, Seiradake E, Dobbie IM, Jones EY, Davis I, Hagen C, Grünewald K. 2014 Super-resolution microscopy using standard fluorescent proteins in intact cells under cryo-conditions. *Nano Lett.* **14**, 4171–4175. (doi:10.1021/nl501870p)
83. Hoffman DP *et al.* 2020 Correlative three-dimensional super-resolution and block-face electron microscopy of whole vitreously frozen cells. *Science* **367**, eaaz5357. (doi:10.1126/science.aaz5357)
84. Wolff G, Hagen C, Grünewald K, Kaufmann R. 2016 Towards correlative super-resolution fluorescence and electron cryo-microscopy. *Biol. Cell* **108**, 245–258. (doi:10.1111/boc.201600008)
85. Xu CS *et al.* 2017 Enhanced FIB-SEM systems for large-volume 3D imaging. *Elife* **6**, e25916. (doi:10.7554/eLife.25916)
86. Jungmann R, Avendaño MS, Woehrstein JB, Dai M, Shih WM, Yin P. 2014 Multiplexed 3D cellular super-resolution imaging with DNA-PAINT and Exchange-PAINT. *Nat. Methods* **11**, 313–318. (doi:10.1038/nmeth.2835)
87. Grimm JB *et al.* 2015 A general method to improve fluorophores for live-cell and single-molecule microscopy. *Nat. Methods* **12**, 244–250. (doi:10.1038/nmeth.3256)
88. Liu Z, Lavis LD, Betzig E. 2015 Imaging live-cell dynamics and structure at the single-molecule level. *Mol. Cell* **58**, 644–659. (doi:10.1016/j.molcel.2015.02.033)
89. Fridy PC *et al.* 2014 A robust pipeline for rapid production of versatile nanobody repertoires. *Nat. Methods* **11**, 1253–1260. (doi:10.1038/nmeth.3170)
90. Oleksiievets N *et al.* 2022 Fluorescence lifetime DNA-PAINT for multiplexed super-resolution imaging of cells. *Commun. Biol.* **5**, 1–8. (doi:10.1038/s42003-021-02976-4)
91. Lin G, Baker MA, Hong M, Jin D. 2018 The quest for optical multiplexing in bio-discoveries. *Chem* **4**, 997–1021. (doi:10.1016/j.chempr.2018.01.009)
92. Helle ØI, Dullo FT, Lahrberg M, Tinguely J-C, Hellesø OG, Ahluwalia BS. 2020 Structured illumination microscopy using a photonic chip. *Nat. Photonics* **14**, 431–438. (doi:10.1038/s41566-020-0620-2)
93. Diederich B, Lachmann R, Carlstedt S, Marsikova B, Wang H, Uwurukundo X, Mosig AS, Heintzmann R. 2020 A versatile and customizable low-cost 3D-printed open standard for microscopic imaging. *Nat. Commun.* **11**, 5979. (doi:10.1038/s41467-020-19447-9)
94. Boland MA, Cohen EA, Flaxman SR, Neil MA. 2021 Improving axial resolution in Structured Illumination Microscopy using deep learning. *Phil. Trans. R. Soc. A* **379**, 20200298. (doi:10.1098/rsta.2020.0298)
95. Gong H, Guo W, Neil MA. 2021 GPU-accelerated real-time reconstruction in Python of three-dimensional datasets from structured illumination microscopy with hexagonal patterns. *Phil. Trans. R. Soc. A* **379**, 20200162. (doi:10.1098/rsta.2020.0162)
96. Phillips MA *et al.* 2020 Cryosim: super-resolution 3d structured illumination cryogenic fluorescence microscopy for correlated ultrastructural imaging. *Optica* **7**, 802–812. (doi:10.1364/OPTICA.393203)
97. Huang X *et al.* 2018 Fast, long-term, super-resolution imaging with hessian structured illumination microscopy. *Nat. Biotechnol.* **36**, 451–459. (doi:10.1038/nbt.4115)

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98. Song L, Lu-Walther H-W, Förster R, Jost A, Kielhorn M, Zhou J, Heintzmann R. 2016 Fast structured illumination microscopy using rolling shutter cameras. *Meas. Sci. Technol.* **27**, 055401. (doi:10.1088/0957-0233/27/5/055401)
99. Appelt D, Ehler E, Mukherjee SS, Heintzmann R, Wicker K. 2022 Polarized illumination coded structured illumination microscopy (picoSIM): experimental results. *Phil. Trans. R. Soc. A* **380**, 20210193. (doi:10.1098/rsta.2021.0193)
100. Mo Y, Feng F, Mao H, Fan J, Chen L. 2021 Structured illumination microscopy artefacts caused by illumination scattering. *Phil. Trans. R. Soc. A* **379**, 20200153. (doi:10.1098/rsta.2020.0153)
101. Ball G, Demmerle J, Kaufmann R, Davis I, Dobbie IM, Schermelleh L. 2015 Simcheck: a toolbox for successful super-resolution structured illumination microscopy. *Sci. Rep.* **5**, 1–12. (doi:10.1038/srep15915)
102. Demmerle J *et al.* 2017 Strategic and practical guidelines for successful structured illumination microscopy. *Nat. Protoc.* **12**, 988–1010. (doi:10.1038/nprot.2017.019)
103. Prakash K, Diederich B, Reichelt S, Heintzmann R, Schermelleh L. 2021 Super-resolution structured illumination microscopy: past, present and future. *Phil. Trans. R. Soc. A* **379**, 20200143. (doi:10.1098/rsta.2020.0143)
104. Heintzmann R. 2021 Answers to fundamental questions in superresolution microscopy. *Phil. Trans. R. Soc. A* **379**, 20210105. (doi:10.1098/rsta.2021.0105)
105. Manton JD. 2022 Answering some questions about structured illumination microscopy. *Phil. Trans. R. Soc. A* **380**, 20210109. (doi:10.1098/rsta.2021.0109)
106. Roth S, Sheppard CJ, Wicker K, Heintzmann, R. 2013 Optical photon reassignment microscopy (OPRA). *Optical Nanoscopy* **2**, 1–6.
107. De Luca GM *et al.* 2013 Re-scan confocal microscopy: scanning twice for better resolution. *Biomedical optics express* **4**, 2644–2656.
108. Miron E *et al.* 2020 Chromatin arranges in chains of mesoscale domains with nanoscale functional topography independent of cohesin. *Science advances* **6**, eaba8811. (doi:10.1126/sciadv.aba8811)
109. Bintu B, Mateo LJ, Su JH, Sinnott-Armstrong NA, Parker M, Kinrot S, Yamaya K, Boettiger AN, Zhuang X. 2018 Super-resolution chromatin tracing reveals domains and cooperative interactions in single cells. *Science*, **362**. (doi:10.1126/science.aau1783)
110. Kirmes I *et al.* 2015 A transient ischemic environment induces reversible compaction of chromatin. *Genome biology* **16**, 1–19. (doi:10.1186/s13059-015-0802-2)
111. Ricci MA, Manzo C, García-Parajo MF, Lakadamyali M, Cosma MP. 2015 Chromatin fibers are formed by heterogeneous groups of nucleosomes in vivo. *Cell* **160**, 1145–1158. (doi:10.1016/j.cell.2015.01.054)
112. Szabo Q *et al.* 2018 TADs are 3D structural units of higher-order chromosome organization in Drosophila. *Science advances* **4**, eaar8082. (doi:10.1126/sciadv.aar8082)
113. Prakash K *et al.* 2015 Superresolution imaging reveals structurally distinct periodic patterns of chromatin along pachytene chromosomes. *Proc. Natl Acad. Sci. USA* **112**, 14 635–14 640. (doi:10.1073/pnas.1516928112)
114. Ochs F *et al.* 2019 Stabilization of chromatin topology safeguards genome integrity. *Nature* **574**, 571–574. (doi:10.1038/s41586-019-1659-4)
115. Rodermund L *et al.* 2021 Time-resolved structured illumination microscopy reveals key principles of Xist RNA spreading. *Science* **372**. (doi:10.1126/science.abe7500)
116. Bond C, Santiago-Ruiz AN, Tang Q, Lakadamyali M. 2022 Technological advances in super-resolution microscopy to study cellular processes. *Molecular Cell* **82**, 315–332. (doi:10.1016/j.molcel.2021.12.022)
117. Schock F, Best G, Celik N, Heintzmann R, Dithmar S, Cremer C. 2022 Structured illumination ophthalmoscope: super-resolution microscopy on the living human eye. *Phil. Trans. R. Soc. A* **380**, 20210151. (doi:10.1098/rsta.2021.0151)
118. Jabermoradi A, Yang S, Gobes MI, van Duynhoven JPM, Hohlbein J. 2022 Enabling single-molecule localization microscopy in turbid food emulsions. *Phil. Trans. R. Soc. A* **380**, 20200164. (doi:10.1098/rsta.2020.0164)
119. Collins JT *et al.* 2020 Robotic microscopy for everyone: the OpenFlexure microscope. *Biomed. Opt. Express* **11**, 2447–2460. (doi:10.1364/BOE.385729)

120. Schindelin J *et al.* 2012 Fiji: an open-source platform for biological-image analysis. *Nat. Methods* **9**, 676–682. (doi:10.1038/nmeth.2019)
121. Diederich B, Then P, Jügler A, Förster R, Heintzmann R. 2019 cellSTORM—Cost-effective super-resolution on a cellphone using dSTORM. *PLoS ONE* **14**, e0209827. (doi:10.1371/journal.pone.0209827)
122. Wang H, Lachmann R, Marsikova B, Heintzmann R, Diederich B. 2022 UCsim2: 2D structured illumination microscopy using UC2. *bioRxiv*. (doi:10.1101/2021.01.08.425840)
123. Prakash K. 2021 Laser-free super-resolution microscopy. *Phil. Trans. R. Soc. A* **379**, 20200144. (doi:10.1098/rsta.2020.0144)
124. Pilger C, Pospíšil J, Müller M, Ruoff M, Schütte M, Spiecker H, Huser T. 2021 Super-resolution fluorescence microscopy by line-scanning with an unmodified two-photon microscope. *Phil. Trans. R. Soc. A* **379**, 20200300. (doi:10.1098/rsta.2020.0300)
125. McFadden D, Amos B, Heintzmann R. 2022 Quality control of image sensors using gaseous tritium light sources. (doi:10.22443/rms.elmi2021.71)
126. Boehm U *et al.* 2021 QUAREP-LiMi: a community endeavor to advance quality assessment and reproducibility in light microscopy. *Nat. Methods* **18**, 1423–1426. (doi:10.1038/s41592-021-01162-y)
127. Hohlbein J, Diederich B, Marsikova B, Reynaud EG, Holden S, Jahr W, Haase R, Prakash K. 2021 Open microscopy in the life sciences: Quo Vadis? arXiv preprint arXiv:2110.13951.

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